WOWII: A counterexample to conjecture 64

December 12, 2025

1 Conjecture

If G is a simple connected graph, then $f(G) \geq \left\lceil \sqrt{a(G) * (1 + (n \bmod D(G)))} \right\rceil$

Conjecture reference.

Notations

- $f(G)$: The number of vertices of a largest induced forest of the graph.
- $a(G)$: The maximum number of vertices such that no two vertices are adjacent.
- $D(G)$: The maximum degree of a graph.

2 Results

2.1 Graph identification

The graph is defined by the following graph6 string:

```
S~~~~~~~w_@@@B?b_Hw@^?F}?NyO^x?^{
```

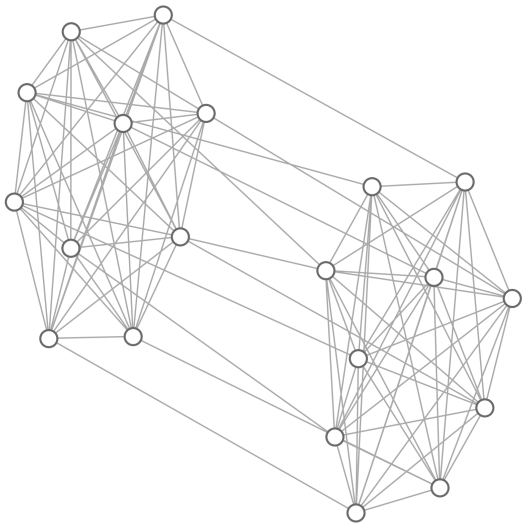


Figure 1: Counterexample found

2.2 Calculated properties

Property	Value
Number of vertices n	20
Number of edges $\lvert E(G)\rvert$	101
Maximum degree $D(G)$	11
Independence number $a(G)$	2
Forest number $f(G)$	4
Size of the minimum Feedback Vertex Set $\nu(G)$	16

Table 1: Properties of the counterexample

2.3 Verification of the inequality

1. Calculation of the left-hand side (LHS): The number of vertices of a largest induced forest of the graph is given as $f(G) = 4$.

$$\text{LHS} = f(G) = 4$$

2. Calculation of the right-hand side (RHS): With $n = 20$, $D(G) = 11$, and $a(G) = 2$, we calculate $n \bmod D(G) = 20 \bmod 11 = 9$. The inner term is $a(G) \cdot (1 + 9) = 2 \cdot 10 = 20$.

$$\text{RHS} = \left\lceil \sqrt{20} \right\rceil = 5$$

Verification :

$$4 \geq 5 \quad \Longrightarrow \quad \textbf{FALSE}$$

3 Conclusion

The conjecture states that $f(G) \geq \left\lceil \sqrt{a(G) * (1 + (n \bmod D(G)))} \right\rceil$. However, we observe:

$$f(G) = 4 \quad \text{and} \quad \text{RHS} = 5$$

$$4 < 5$$

The inequality is not satisfied. **This graph is therefore a valid counterexample.**